

A055 — Blind Topological Particle Spectrum Falsifier Signed-Traveling-Wave Certification Protocol

SST-Workbench

Version v0.3.1 — SST Falsifier Framework v1.0.4

1 Identification, scope, and status

Field	Value
Catalog ID	A055
Version	v0.3.1
Framework	SST Falsifier Framework v1.0.4, CANONICAL_FROZEN
Profile	CPU: Python FP64 reference plus C++17/OpenMP FP64 certification
Scientific status	Frozen falsification protocol; physical mechanism not established by implementation validation alone

This version is a method-corrected reanalysis of sealed finite-core vortex-filament campaigns. It does not modify the upstream dynamics and does not claim an independent physical replication.

2 Research question and objective

2.1 Single falsifiable research question

After excluding the automatic complex-conjugate partner of a real projected Jacobian, does any anonymous three-component vortex compound contain two *distinct positive-temporal-frequency* Kelvin-dominated modes with opposite signed spatial propagation, matched in frequency under the frozen threshold, and recurring in at least two circulation sectors?

2.2 Scientific objective

The objective is to replace an algebraically non-discriminating frequency-sign test by a spatial propagation observable. The test reconstructs complex eigenfields in physical space, measures their signed arclength-Fourier content, and then asks whether opposite directions coexist as distinct positive-frequency branches. Linear mode evidence is kept separate from spatial convergence, relative-periodic-orbit (RPO) recurrence, and conditional Floquet evidence.

3 Competing hypotheses and allowed conclusions

Null hypothesis H_0 . No anonymous compound satisfies the full signed-travel pair rule in at least two circulation sectors.

Alternative hypothesis H_1 . At least one anonymous compound satisfies the rule in at least two circulation sectors. This establishes only a projected linear traveling-wave sector. Mechanism promotion requires the same sector to satisfy the sealed upstream spatial-convergence and nonlinear RPO gates.

Falsification semantics. If H_0 is retained at G3, the corrected counterpropagating mechanism is not observed in the tested ensemble. If G3 passes but G8 fails, a linear signed-travel signal exists without nonlinear mechanism certification. G2 or G5 failure is numerical/backend invalidity, not physical evidence.

4 Assumptions, domain of validity, and known edge cases

- A1. The upstream projected Jacobian and finite mode basis are admissible diagnostics only within the registered finite-core filament model.
- A2. The local normal/binormal frame and arclength phase are deterministic numerical gauges; conclusions are limited to this frozen construction.
- A3. Positive temporal frequency selects one representative from each real-matrix complex-conjugate pair.
- A4. Historical isolated-centerline results and the compound results share methodology and are not treated as independent physical samples.

Known limits. Near-zero temporal frequencies are excluded by a relative floor. Modes with weak tested-harmonic participation or weak directional purity remain unclassified rather than being forced into a direction. No continuum-Euler theorem follows from the finite projected spectrum.

5 Symbols, units, and dimensional consistency

Symbol	Meaning	SI unit	Dimensions
λ_j	projected-Jacobian eigenvalue	1	1
ω_j	$\text{Im } \lambda_j$ in the upstream nondimensional time	1	1
$P_{j,k}^\pm$	signed spatial harmonic power	1	1
$\chi_{j,k}$	signed traveling purity	1	1
$\eta_{j,k}$	harmonic participation	1	1
A_ω	relative opposite-direction frequency asymmetry	1	1

All decision variables are ratios or normalized projected quantities and are dimensionless. No SST dimensional constant enters the blind directional statistic.

6 Exact mathematical model and observables

For component c , the reconstructed complex eigenfield is projected on a deterministic periodic Bishop parallel-transport transverse frame with the closed-loop holonomy distributed continuously over arclength to obtain complex scalar series $a_{n,c}(s)$ and $a_{b,c}(s)$. For arclength phase $\theta_c(s) \in [0, 2\pi)$ and harmonic k ,

$$P_{j,k}^\pm = \sum_c (|\hat{a}_{n,c}(\pm k)|^2 + |\hat{a}_{b,c}(\pm k)|^2), \quad (1)$$

with Fourier coefficients computed directly at the arclength phases. Directional purity is

$$\chi_{j,k} = \frac{P_{j,k}^+ - P_{j,k}^-}{P_{j,k}^+ + P_{j,k}^-}, \quad (2)$$

and harmonic participation is

$$\eta_{j,k} = \frac{P_{j,k}^+ + P_{j,k}^-}{P_{j,\perp}}. \quad (3)$$

Two distinct positive-frequency modes of opposite spatial direction at the same k are frequency matched by

$$A_\omega = \frac{|\omega_1 - \omega_2|}{(\omega_1 + \omega_2)/2}. \quad (4)$$

Because the upstream projected basis is orthonormal and carries frozen family labels, the Kelvin-family coefficient fraction is

$$\zeta_j = \frac{\sum_{i \in \mathcal{K}} |v_{ij}|^2}{\sum_i |v_{ij}|^2}. \quad (5)$$

The conjugate-invariant physical direction is equivalently

$$d_{j,k} = \text{sgn}(\omega_j \chi_{j,k}). \quad (6)$$

Under conjugation, $\omega \mapsto -\omega$ and $\chi \mapsto -\chi$, so d is unchanged. The implementation uses only the $\omega > 0$ representative; an automatic conjugate partner therefore cannot create the opposite direction.

6.1 Frozen thresholds

A traveling candidate must satisfy $\zeta \geq 0.50$, $\eta \geq 0.50$, $|\chi| \geq 0.60$, and $|\text{Re } \lambda|/|\text{Im } \lambda| \leq 1.0$. A bidirectional pair requires $A_\omega \leq 0.25$. An anonymous candidate is a blind discovery only if a pair appears in at least two registered circulation sectors. Fine-resolution eigenvalue recomputation must match the sealed stored multiset within relative assignment error 10^{-8} .

6.2 Machine-readable equation/step register

Machine-readable science contract

Research question After removing the conjugate-eigenvalue artifact, do sealed three-component vortex compounds exhibit at least two distinct positive-temporal-frequency Kelvin-dominated modes with opposite signed spatial propagation and preregistered frequency matching?

Objective Reanalyse sealed upstream compound dynamics with a propagation-direction observable derived from the spatial Fourier content of reconstructed complex eigenfields, retain the prior isolated-centerline campaign only as an anonymous control baseline, and separate linear traveling-wave evidence from spatial convergence, RPO and Floquet promotion.

H_0 H0: no anonymous compound has a frequency-matched bidirectional traveling pair that survives the frozen harmonic-participation, traveling-purity and oscillatory-quality thresholds in at least two circulation sectors.

H_1 H1: at least one anonymous compound has such a pair in at least two circulation sectors; physical mechanism promotion additionally requires the same sector to satisfy the sealed upstream spatial-convergence and nonlinear RPO conditions.

Registered equations

E1: Spatial signed-harmonic powers

$$P_{j,k}^\pm = \sum_c (|\hat{a}_{n,c}(\pm k)|^2 + |\hat{a}_{b,c}(\pm k)|^2) \quad (7)$$

Dimensional check: The eigenvectors are normalized perturbation coordinates, so Fourier amplitudes and their squared powers are dimensionless.

E2: Traveling purity

$$\chi_{j,k} = \frac{P_{j,k}^+ - P_{j,k}^-}{P_{j,k}^+ + P_{j,k}^-} \quad (8)$$

Dimensional check: Ratio of equal-dimension powers; chi is dimensionless and lies in [-1,1].

E3: Harmonic participation

$$\eta_{j,k} = \frac{P_{j,k}^+ + P_{j,k}^-}{P_{j,\perp}} \quad (9)$$

Dimensional check: Ratio of powers; eta is dimensionless.

E4: Frequency matching of opposite spatial directions

$$A_\omega = \frac{|\omega_1 - \omega_2|}{(\omega_1 + \omega_2)/2} \quad (10)$$

Dimensional check: Frequency difference divided by frequency; dimensionless.

E5: Conjugate-invariant physical direction

$$d_{j,k} = \text{sgn}(\omega_j \chi_{j,k}) \quad (11)$$

Dimensional check: Sign of a dimensionless product; d is dimensionless.

E6: Kelvin-family coefficient fraction

$$\zeta_j = \frac{\sum_{i \in \mathcal{K}} |v_{ij}|^2}{\sum_i |v_{ij}|^2} \quad (12)$$

Dimensional check: Ratio of squared dimensionless coordinates in an orthonormal projected basis; zeta is dimensionless.

Registered code-to-mathematics steps

1. **S1** [G1; equations E5]: Resolve and hash the sealed upstream compound campaign and the completed anonymous isolated-control campaign; verify both seals before reading scientific records.
2. **S2** [G2; equations E1]: Recompute the fine-resolution upstream projected Jacobian from sealed blind inputs and require eigenvalue-multiset agreement with the stored sealed spectrum.
3. **S3** [G2; equations E1, E2, E3, E5, E6]: Reconstruct each positive-temporal-frequency complex eigenfield, compute its Kelvin-family coefficient fraction, and compute signed spatial harmonic powers, purity and participation in a periodic holonomy-corrected Bishop frame for the pre-registered harmonics.
4. **S4** [G3; equations E2, E3, E4, E5, E6]: Apply the target-free bidirectional-pair rule and require recurrence in at least two circulation sectors of an anonymous candidate.
5. **S5** [G5; equations E1]: Certify the signed-harmonic power calculation against the instance-local C++17/OpenMP FP64 implementation without changing the Python reference result.

6. **S6** [G8; equations E5]: Require upstream spatial convergence and accepted nonlinear RPO in the same sector before mechanism promotion; Floquet remains conditional on the upstream RPO gate.
7. **S7** [G9; equations E4, E5]: After framework reveal verification only, attach semantic architecture/component labels and perform paired, not pooled, architecture and opposed-component contrasts.

7 Exact algorithm: equation-to-code mapping

- S1. Verify the framework-frozen protocol and both sealed upstream campaigns without reading semantic private mappings.
- S2. Load anonymous compound geometry, resample to the sealed finest resolution, reconstruct the exact upstream mode basis, and recompute the projected Jacobian using the isolated upstream blind runner.
- S3. Compare the recomputed eigenvalue multiset with the stored sealed spectrum using minimum-cost assignment normalized by spectral scale.
- S4. For each positive-frequency eigenmode, reconstruct the physical complex field in the periodic Bishop frame, evaluate Eqs. (1)–(3), and require Eq. (5) for the registered harmonics.
- S5. Form opposite-direction pairs only from distinct positive-frequency eigenmodes and apply Eq. (4).
- S6. Apply the anonymous sector-recurrence discovery rule.
- S7. Certify the signed-harmonic power implementation with the instance-local C++17/OpenMP FP64 kernel against Python FP64.
- S8. Promote a mechanism only when a pair-bearing sector also has sealed upstream spatial convergence and accepted RPO; Floquet remains conditional on that upstream RPO.
- S9. After framework reveal commitment verification, attach semantic labels and compute paired reveal contrasts; never retune blind thresholds.

8 Source, provenance, and independence contract

Two sealed simulation sources are admissible: the authoritative three-component compound campaign and the completed historical isolated-centerline campaign. Both are hashed by the framework source manifest. The latter is a control baseline only, not an independent replication. A missing or invalid seal makes G1 unresolved. v0.3.1 never launches a long upstream campaign automatically.

Exclusion rules. Unsealed, partial, differently configured, or ambiguous upstream campaigns are excluded before scoring. A recomputation/backend mismatch is numerical invalidity. Semantic private mappings are excluded from blind computation.

9 Gate DAG and decision semantics

Gate	Question	PASS		FAIL / other	
G0	Frozen protocol intact?	framework	verifica-	protocol	mismatch
		tion		aborts	

G1	Sealed upstreams admissible?	both seals valid	missing/invalid is UNRESOLVED
G2	Reference numerics valid?	analytic selftest and eigenvalue recomputation parity pass	numerical FAIL/UNRESOLVED
G3	Anonymous signed-travel discovery?	at least one candidate passes in at least two sectors	FAIL if none
G4	Independent confirmation?	disabled	NOT_APPLICABLE
G5	Native CPU metric parity?	C++ FP64 relative $L2 \leq 10^{-10}$	FAIL/UNRESOLVED
G6	GPU screening?	disabled	NOT_APPLICABLE
G7	Independent physical replication?	disabled	NOT_APPLICABLE
G8	Nonlinear mechanism promotion?	same pair-bearing sector also spatial-converged and RPO-accepted	FAIL if none after G3/G5 pass
G9	Reveal integrity?	framework commitments verify; semantic analysis runs afterward	framework-controlled

10 Numerical certification, convergence, and error budget

The upstream resolution ladder, finite-core closure, Jacobian finite-difference steps, spatial-convergence gates, nonlinear integration and RPO/Floquet definitions are not changed by A055 v0.3.1. The new analysis recomputes only the sealed finest-resolution projected Jacobian needed to recover eigenvectors. Minimum-cost eigenvalue assignment to the sealed spectrum must have maximum relative error $\leq 10^{-8}$. This prevents a changed runner/backend from silently entering the new statistic.

The spatial Fourier projection is evaluated directly at arclength phase rather than assuming uniform vertex index. A periodic Bishop frame is obtained by minimal tangent-to-tangent parallel transport followed by a distributed closed-loop holonomy correction; this avoids Frenet-frame sign flips near low curvature. A candidate must also carry at least half of its orthonormal projected coefficient power in the registered Kelvin family. The C++ certification uses nonuniform deterministic phase grids and mixed signed-harmonic complex signals.

11 Backend authority and parity

Python/NumPy FP64 is the reference implementation of the new signed-harmonic statistic. The instance-local C++17/OpenMP FP64 kernel is certification authority for Eq. (1); the framework records compiler, ABI, flags and build fingerprint. No Python fallback may satisfy G5. The upstream compound Jacobian must use the same backend class recorded by its sealed backend qualification in a production FULL/CERTIFY run.

12 Blindness and reveal boundary

Target-bearing semantic architecture/component labels and historical identity hypotheses are confined to the framework-private directory and protected by nonced commitments. The blind computation sees anonymous candidate IDs, registered circulation-sector labels, geometry, sealed numerical records and the frozen thresholds only. The framework scans the public tree for forbidden private terms before BLIND packaging.

13 Statistics and multiplicity

G3 is an existence/recurrence mechanism gate, not a population-significance claim: a candidate must reproduce the pair in at least two sectors. Reveal-only architecture comparisons use matched assignment/provider-stratum/sector cells. Binary paired contrasts report the discordant counts and exact McNemar/binomial result. Opposed-component comparisons are performed within candidate before aggregation. Pooled unpaired medians are descriptive only and are not the primary identity statistic.

14 Results insertion contract

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	42d9419272431e6b43171e3a0cc67c3ed2d2700d6a16f76b5c05c69f5f395ace
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	84c69a15e3335f4673a5c649b9a6dfbd6f6f5699ab3790d50ad46fdd9381c8fe

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen framework protocol intact?
G1	PASS	Are the sealed anonymous compound and isolated-control upstream c
G2	PASS	Does fine-resolution Jacobian recomputation reproduce the sealed spe
G3	FAIL	Does any anonymous compound exhibit a qualified frequency-matched
G4	NOT_APPLICABLE	Is there an independent held-out physical confirmation source?
G5	PASS	Does strict instance-local C++ FP64 certification reproduce the Pyth
G6	NOT_APPLICABLE	Is optional GPU screening enabled and parity-qualified?
G7	NOT_APPLICABLE	Is independent cross-source physical replication available?
G8	NOT_RUN_PREREQUISITE	Does at least one blind pair-bearing sector also satisfy sealed upstream
G9	PASS	After all blind gates, do reveal commitments verify and paired semantic

Backend manifest

```
{
  "a054_backend": {
    "actual": "cpp_pybind11_openmp",
    "sealed": "cpp_pybind11_openmp"
  },
  "framework_version": "1.0.4",
  "instance_cpp_metric": {
    "actual_backend": "openmp",
    "authority": "CERTIFICATION",
    "build": {
      "actual_backend": "openmp",
      "artifact": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055_Blind",
      "backend_info": {
```

```

    "backend": "openmp",
    "kernel": "biot_savart_plus_signed_harmonic_powers",
    "openmp_compiled": true,
    "openmp_max_threads": 1,
    "precision": "float64"
  },
  "compiler": "C:\\Program Files\\Microsoft Visual Studio\\2022\\Community\\VC\\Tools\\MSVC",
  "compiler_family": "msvc",
  "compiler_version": "Microsoft (R) C/C++ Optimizing Compiler Version 19.44.35228 for x64",
  "error": null,
  "fingerprint_sha256": "3b80d88cce4a2480b15634854b30e989dbcdaba60ab09a4252c38ca764720f73",
  "flags": [
    "C++17",
    "O2/O3",
    "OpenMP"
  ],
  "python_abi": "cp314-win_amd64",
  "success": true,
  "toolset": "14.44.35207"
},
"parity": {
  "candidate_norm": 139.21695630910853,
  "pass": true,
  "reference_norm": 139.21695630910853,
  "relative_l2": 2.1516832794566584e-16,
  "tolerance": 1e-10
},
"status": "PASS"
},
"schema": "AO55-BACKEND-MANIFEST-031",
"summary": {
  "framework_version": "1.0.4",
  "interpretation": "A signed traveling pair is a linear projected-mode diagnostic only; phys",
  "legacy_control": {
    "accepted_rpo": 0,
    "particle_promotions": 0,
    "qualified_modes": 0,
    "unique_mode_resolution_cells": 920
  },
  "mode": "FULL",
  "schema": "AO55-BLIND-SUMMARY-031",
  "signed_travel": {
    "mechanism_promoted_sector_count": 0,
    "robust_candidate_count": 0,
    "row_count": 288
  }
}
}
}

```

Source manifest

```

{
  "manifest_sha256": "804e3c8a5814c36375694cd27c53bceb321cf40e0e31dd4fb90f9441d6bb7e3b",

```

```

"schema": "SST-SOURCE-MANIFEST-2",
"sources": [
  {
    "catalog_id": "A054",
    "directory_manifest_sha256": "d3e5dc9bbc03bb4e59fe99602381dc2e6797ae4729837c5d46ac0b46777",
    "evidence_class": "simulation",
    "exists": true,
    "file_count": 170,
    "independence_family": "A054_FINITE_CORE_COMPOUND",
    "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A054_nucleon",
    "provenance_family": "A054_V020_R2_SEALED",
    "registry_hint": {
      "catalog_id": "A054",
      "hits": []
    },
    "role": "discovery",
    "source_id": "A054_COMPOUND",
    "version": "v0.2.0-r2"
  },
  {
    "catalog_id": "A055",
    "directory_manifest_sha256": "546eacb9496120b8e12d58f0f3884e16c9f395c084d2ab9f482b1cf1020",
    "evidence_class": "simulation",
    "exists": true,
    "file_count": 114,
    "independence_family": "A055_V030_ISOLATED_CONTROL",
    "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055_Blind_T",
    "provenance_family": "A055_V030_SEALED",
    "registry_hint": {
      "catalog_id": "A055",
      "hits": []
    },
    "role": "implementation_control",
    "source_id": "A055_V030_CONTROL",
    "version": "v0.3.0"
  }
],
"workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

A055 v0.3.1 post-run discussion

The reveal phase attaches semantic labels only after framework commitment verification. The primary architecture comparisons are paired by component assignment, provider stratum, and circulation sector; pooled unpaired medians are not used as the primary identity statistic.

```

{
  "a054_private_mapping_commitment_verified": true,
  "architecture_paired_contrasts": {
    "B": {
      "both": 0,
      "control_only": 0,
      "linked_only": 0,

```

```

    "mcnemar_exact_p": null,
    "mean_pair_indicator_delta_linked_minus_unlinked": 0.0,
    "median_best_frequency_asymmetry_delta": null,
    "n_pairs": 96,
    "neither": 96
  },
  "G": {
    "both": 0,
    "control_only": 0,
    "linked_only": 0,
    "mcnemar_exact_p": null,
    "mean_pair_indicator_delta_linked_minus_unlinked": 0.0,
    "median_best_frequency_asymmetry_delta": null,
    "n_pairs": 96,
    "neither": 96
  }
},
"interpretation_guard": "All architecture and component comparisons are reveal-only and paired",
"legacy_isolated_controls": {
  "5_2": {
    "accepted_rpo": 0,
    "case_count": 1,
    "guard": "Legacy +/- imaginary-frequency availability is not physical counterpropagation.",
    "legacy_oscillatory_available": 33,
    "provider_runs": 2,
    "qualified_modes": 0,
    "unique_cells": 40
  },
  "6_1": {
    "accepted_rpo": 0,
    "case_count": 1,
    "guard": "Legacy +/- imaginary-frequency availability is not physical counterpropagation.",
    "legacy_oscillatory_available": 37,
    "provider_runs": 2,
    "qualified_modes": 0,
    "unique_cells": 40
  }
},
"opposed_component_paired_contrasts": {
  "B": {
    "fraction_6_1_higher": 0.0,
    "mean_pair_indicator_delta_6_1_minus_5_2": 0.0,
    "median_frequency_asymmetry_delta_6_1_minus_5_2": null,
    "median_pair_indicator_delta_6_1_minus_5_2": 0.0,
    "paired_candidate_count": 24
  },
  "G": {
    "fraction_6_1_higher": 0.0,
    "mean_pair_indicator_delta_6_1_minus_5_2": 0.0,
    "median_frequency_asymmetry_delta_6_1_minus_5_2": null,
    "median_pair_indicator_delta_6_1_minus_5_2": 0.0,
    "paired_candidate_count": 24
  }
},

```

```

    "U": {
      "fraction_6_1_higher": 0.0,
      "mean_pair_indicator_delta_6_1_minus_5_2": 0.0,
      "median_frequency_asymmetry_delta_6_1_minus_5_2": null,
      "median_pair_indicator_delta_6_1_minus_5_2": 0.0,
      "paired_candidate_count": 24
    }
  },
  "polarity_paired_contrasts": {
    "B": {
      "candidate_count": 24,
      "mean_delta": 0.0,
      "median_opposed_pair_fraction_minus_q0": 0.0,
      "opposed_higher_fraction": 0.0
    },
    "G": {
      "candidate_count": 24,
      "mean_delta": 0.0,
      "median_opposed_pair_fraction_minus_q0": 0.0,
      "opposed_higher_fraction": 0.0
    },
    "U": {
      "candidate_count": 24,
      "mean_delta": 0.0,
      "median_opposed_pair_fraction_minus_q0": 0.0,
      "opposed_higher_fraction": 0.0
    }
  },
  "schema": "A055-REVEALED-ANALYSIS-031",
  "working_hypotheses": {
    "neutron_like": {
      "mechanism_promoted_fraction": 0.0,
      "n": 24,
      "rpo_fraction": 0.0,
      "signed_pair_fraction": 0.0,
      "spatial_converged_fraction": 0.0
    },
    "proton_like": {
      "mechanism_promoted_fraction": 0.0,
      "n": 12,
      "rpo_fraction": 0.0,
      "signed_pair_fraction": 0.0,
      "spatial_converged_fraction": 0.0
    }
  }
}

```

The authoritative values remain the machine-readable gate ledger, signed-travel JSON, backend manifest, source manifest and output manifest.

15 Allowed interpretation

A G3 PASS means a genuine signed spatial bidirectional sector exists in the finite projected model under the frozen thresholds; it does not identify a physical particle. A G8 PASS is stronger but still restricted to the finite-core filament model. G8 FAIL after G3 PASS means the linear traveling sector did not acquire the registered nonlinear confinement evidence. G3 FAIL falsifies the corrected traveling-pair claim for the tested anonymous ensemble. Numerical or source failures remain unresolved rather than being converted into physical negatives.

16 Reproducibility and exact commands

Canonical Workbench location:

```
01_research\A_falsifiers\A055_Blind_Topological_Particle_Spectrum\A055-v0.3.1
```

Canonical framework location:

```
06_templates\SST_Falsifier_Framework\SST_Falsifier_Framework_v1.0.4
```

Recommended production command:

```
run_all.cmd full C:\workspace\projects\SST-Workbench
```

Blind-only command:

```
run_blind_only.cmd FULL C:\workspace\projects\SST-Workbench
```

The instance is shipped already frozen. Re-running FREEZE verifies byte identity and refuses protocol mutation under the same version.

References

- [1] SST-Workbench A054 v0.2.0-r2, sealed finite-core three-component dynamical certification campaign and isolated blind runner, project software and outputs.
- [2] SST-Workbench A055 v0.3.0, completed blind topological particle-spectrum campaign, project software and outputs.
- [3] SST Falsifier Framework v1.0.4, CANONICAL_FROZEN, 7 October 2026.